

Setup & Usage Guide

Overview

This trading bot places **Market**, **Limit**, and **Stop-Market** orders on the **Binance Futures Testnet (USDT-M)** via a clean CLI. An optional **natural-language (--chat) mode** powered by **OpenRouter** (**openai/gpt-oss-120b:free**) lets you describe trades in plain English and have the LLM extract the order parameters automatically.

■■ All orders target the **Testnet** only. No real money is ever at risk. The base URL is hardcoded to <https://testnet.binancefuture.com>

Prerequisites

- Python 3.8 or higher
- pip (comes with Python)
- A Binance Futures Testnet account (free)
- An OpenRouter account — *only* if you plan to use **--chat** mode
- Git (optional, for uploading to GitHub)

Project Structure

```
trading_bot/
├── bot/
│   ├── __init__.py
│   ├── config.py          ← constants & OpenRouter settings
│   ├── client.py          ← Binance REST client (HMAC-SHA256 signing)
│   ├── orders.py          ← Market / Limit / Stop-Market logic
│   ├── validators.py       ← input validation
│   ├── logging_config.py  ← file + console logging
│   └── llm_parser.py       ← OpenRouter LLM natural-language parser
├── logs/
│   ├── market_order_sample.log
│   └── limit_order_sample.log
├── cli.py                  ← CLI entry point
├── test_bot.py             ← 22 offline unit tests
├── requirements.txt
├── .env.example
├── .gitignore
└── README.md
```

Setup Steps

1 • Get Binance Futures Testnet API Keys

Step 1	Visit testnet	Go to https://testnet.binancefuture.com
Step 2	Sign in	Click Log In with GitHub (free account)
Step 3	Generate key	Open the top-right profile menu → API Keys → Generate Key
Step 4	Save secrets	Copy API Key and Secret Key immediately — the secret is shown only once

2 • Get OpenRouter API Key (--chat mode only)

Step 1	Register	Go to https://openrouter.ai and sign up
Step 2	Create key	Navigate to Keys in the dashboard → Create Key
Step 3	Copy key	Save the key — it won't be shown again

■ Skip this section entirely if you only want standard CLI mode.

3 • Install the Bot

```
# Unzip the project
unzip trading_bot.zip
cd trading_bot

# (Optional but recommended) create a virtual environment
python -m venv .venv
source .venv/bin/activate      # macOS / Linux
.venv\Scripts\activate        # Windows

# Install dependencies
pip install -r requirements.txt
```

4 • Configure Credentials

```
# Copy the example file
cp .env.example .env

# Open .env in any text editor and fill in:
BINANCE_API_KEY=your_binance_api_key_here
BINANCE_API_SECRET=your_binance_api_secret_here

# Only needed for --chat mode:
OPENROUTER_API_KEY=your_openrouter_api_key_here
```

■ **Never commit your .env file.** It is already listed in .gitignore. Only .env.example (with placeholder values) should be pushed to GitHub.

Running the Bot

Standard Mode (argparse flags)

Pass all order parameters directly as CLI flags:

Command	Description
<code>python cli.py --symbol BTCUSDT --side BUY --type MARKET --amount 0.001</code>	Place a Market BUY order for 0.001 BTC
<code>python cli.py --symbol BTCUSDT --side SELL --type MARKET --amount 0.001</code>	Place a Market SELL order
<code>python cli.py --symbol BTCUSDT --side BUY --type LIMIT --amount 0.001 --price 30000</code>	Place a Limit BUY at \$30,000
<code>python cli.py --symbol ETHUSDT --side SELL --type LIMIT --amount 0.001 --price 2000</code>	Place a Limit SELL for ETH at \$2,000
<code>python cli.py --symbol BTCUSDT --side SELL --type STOP --amount 0.001 --price 28000</code>	Stop-Market SELL triggered at \$28,000 (bonus order type)
<code>python cli.py --help</code>	Show all available flags

Chat / NLP Mode (--chat flag)

Use **--chat** to enter an interactive loop where you describe orders in plain English. The bot sends your text to the OpenRouter LLM, which extracts the structured parameters, validates them, asks for confirmation, then places the order.

■ The LLM always asks for confirmation before any order is placed — you can type 'n' or just press *Enter* to cancel safely.

Example Output

Troubleshooting

Error / Symptom	Cause	Fix
Missing API credentials	.env not configured	Copy .env.example → .env and fill in keys
HTTP 400 — Invalid quantity	Quantity below testnet minimum	Use at least 0.001 for BTCUSDT
HTTP 401 — Unauthorized	Wrong API key/secret	Re-generate keys on testnet dashboard
HTTP 400 — Invalid price	Price too far from market	Use a realistic price near the current market
OPENROUTER_API_KEY not set	that used without key	Add OPENROUTER_API_KEY to .env
LLM returned non-JSON	Model hallucinated	Rephrase your order description and retry
Connection error / timeout	Network issue	Check internet connection; bot retries are not automatic

Submitting to GitHub

```
git init
git add .
git commit -m 'feat: Binance Futures trading bot with LLM chat mode'
git branch -M main
git remote add origin https://github.com/<your-username>/trading-bot.git
git push -u origin main
```

Before pushing: run the bot against the testnet with real keys, then replace the sample log files in **logs/** with your actual output logs to satisfy the deliverable requirement.

■ Submit the GitHub repo URL (not this zip) through the Google Form shared by the hiring team.